CHANNEL 04 · ACTIVE CH-04 · 914 MHz



BUS SYNCED // 914.99 MHz · I2C 400 kHz · PWM 75%

CLASSIFIED — PROBLEM STATEMENT CHANNEL

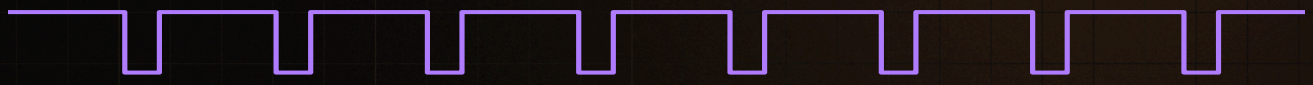
SMART HOME & CONSUMER ELECTRONICS

PROBLEM STATEMENTS · ROUND 1

24-HOUR NATIONAL-LEVEL HARDWARE HACKATHON

29–30 JANUARY 2027 · FR. C. RODRIGUES INSTITUTE OF TECHNOLOGY, VASHI

DOMAIN SLOT 04 CH-04 · 914 MHz



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SMART HOME & CONSUMER ELECTRONICS

The next generation of intelligent, energy-aware and secure consumer appliances. Embed edge intelligence, robust fail-safes and intuitive automation directly into the devices of modern homes.

— AREAS TO EXPLORE

Low-Latency Automation | Smart Energy Metering | Ambient Hazard Sensors | Edge AI Local Hubs | Matter & BLE Mesh

— WHO SHOULD EXPLORE

IoT Architectures · Embedded Microcontrollers (ESP32 / ARM Cortex) · Wireless Mesh & Consumer UX

> *"Bringing intelligence inside the home — turn ordinary appliances into aware, dependable systems."*

— IN THIS CHANNEL

PS-01	Detection of Hidden Water Leakage in Domestic Plumbing Systems
PS-02	Identification of High-Energy-Consuming Household Appliances
PS-03	LPG Leakage Detection with Automatic Gas Supply Isolation
PS-04	Detection of Drinking Water Quality Changes at Household Supply Points
PS-05	Automatic Backup Power Management for Essential Household Appliances

PS-01 DETECTION OF HIDDEN WATER LEAKAGE IN DOMESTIC PLUMBING SYSTEMS

Recommended for First & Second Year students

A · PROBLEM DEFINITION

Leaks inside walls, under floors or in concealed pipes release water slowly and silently. They are noticed only when dampness, stains, mould or structural damage appear. The challenge is to detect small, continuous leaks early using measurements taken at the water inlet or at vulnerable points, and to tell them apart from normal household water use.

B · CONTEXT

Hidden leaks waste a scarce resource, raise water bills and damage walls, flooring and electrical fittings. Conventional detection relies on visible signs or a plumber's inspection after damage has occurred, and breaking walls to trace a leak is costly. Many urban homes and rental apartments need an affordable device that can be fitted without major plumbing work, warns residents early and, if desired, shuts off the supply to limit damage.

C · CORE FEATURES

- Measure water flow and/or pipe pressure at the main inlet to build a normal usage profile.
- Detect continuous low flow during periods of no expected use (e.g. at night) and abnormal pressure decay.
- Sense moisture at vulnerable points such as under sinks, near water heaters and around washing machines.
- Alert residents locally and remotely, with an optional automatic shut-off valve.
- Log daily water consumption and display leak history.

D · DESIGN REQUIREMENTS

- Users: homeowners, tenants, housing societies and small hotels or hostels.
- Installation: retrofittable at the inlet or on exposed pipes with minimal plumbing work and no wall breaking.
- Reliability: few false alarms during legitimate use (filling a tank, running a garden tap); simple setup for non-technical users.
- Sustainability: measurable water savings; low power draw and long battery life.

E · TECHNICAL FOCUS

SENSING Hall-effect or ultrasonic flow sensor, pressure sensor, conductive or capacitive moisture probes. Compare at least two sensing approaches and justify the final choice.

PROCESSING & LOGIC Low-cost microcontroller. Detection may use night-time baseline flow, minimum-flow thresholds or a pressure-decay test with supply closed; thresholds must be justified (e.g. leak rates of 1–10 litres per hour).

COMMUNICATION Buzzer/LED locally, with optional Wi-Fi, Bluetooth or GSM notification.

POWER SUPPLY Battery or mains with backup; state the expected battery life.

SIMULATION Generate 24-hour household water-use profiles with injected leaks of different sizes; show detection time and false-alarm behaviour in Python, MATLAB or equivalent.

OPTIONAL Automatic solenoid shut-off valve, per-fixture leak localisation, tank-overflow detection, ML usage classification.

F · EXPECTED OUTCOME

- System block diagram and sensor placement plan.
- Simulation of normal usage and at least three leak scenarios (different rates) with detection results.
- Detection time and false-alarm analysis for the chosen thresholds.
- Bill of materials with estimated cost and a simple installation plan.
- Estimate of water and money saved for a typical household.

PS-02 IDENTIFICATION OF HIGH-ENERGY-CONSUMING HOUSEHOLD APPLIANCES

Recommended for First & Second Year students

A · PROBLEM DEFINITION

Households normally see only a single monthly electricity bill and cannot tell which appliances (air conditioner, geyser, refrigerator, pump, iron) consume the most energy. The challenge is to identify appliance-level consumption using minimal sensing, either at the main supply (non-intrusive load monitoring) or at socket level, and to present the results in a way that supports energy-saving decisions.

B · CONTEXT

Electricity cost and demand in homes are rising, yet consumers lack insight into where energy actually goes. Smart sub-meters on every appliance are expensive and impractical, while the bill gives no breakdown. A low-cost system that recognises appliances from their electrical behaviour, ranks the biggest consumers and flags abnormal usage can help households cut consumption, plan purchases and understand the effect of their daily habits.

C · CORE FEATURES

- Measure voltage, current, real power, power factor and energy at the chosen measurement point.
- Detect appliance switching events from step changes and start-up transients in power.
- Recognise appliances (or appliance categories) from their electrical signatures.
- Estimate energy and cost per appliance and rank the top consumers.
- Flag unusual behaviour, such as a refrigerator running continuously or a geyser left on.

D · DESIGN REQUIREMENTS

- Users: households, student hostels and small shops.
- Cost: significantly cheaper than a meter on every appliance; fitted at the distribution board or a few plugs.
- Privacy: data processed locally where possible, with no unnecessary upload of usage patterns.
- Accuracy & safety: a justified measurement accuracy (e.g. within ±5%); mains-connected sensing must be isolated and enclosed.

E · TECHNICAL FOCUS

SENSING Current transformer or shunt plus voltage sensor, or a dedicated energy-metering IC; sampling rate adequate to capture switching transients.

PROCESSING & ANALYTICS Microcontroller or single-board computer. Event detection by thresholding power changes, feature extraction (steady power, power factor, start-up spike, duration) and simple classification (rules or nearest-neighbour).

COMMUNICATION Wi-Fi or Bluetooth to a phone or dashboard; local display acceptable.

DATA Published household datasets (e.g. REDD, UK-DALE) or synthetic load profiles for the simulation.

SIMULATION Aggregate-load simulation in Python, MATLAB or equivalent that disaggregates a mixed signal into appliance-level estimates and reports accuracy.

OPTIONAL ML disaggregation, harmonic signatures, tariff-aware advice, appliance-health alerts.

F · EXPECTED OUTCOME

- System architecture covering sensing, event detection and classification.
- Signature table for at least four appliances (power, power factor, transient behaviour).
- Simulation on a 24-hour aggregate load, with identification accuracy (%) and energy-estimation error (%).
- Example user report ranking appliances by energy and cost.
- Cost estimate and limitations (e.g. similar-signature appliances).

PS-03 LPG LEAKAGE DETECTION WITH AUTOMATIC GAS SUPPLY ISOLATION

Recommended for Second & Third Year students

A · PROBLEM DEFINITION

LPG (a propane–butane mixture) is heavier than air and collects near the floor, where it can become explosive at roughly 2–9% concentration in air. Typical household detectors only sound an alarm and depend on the user to react. The challenge is to detect leaks reliably well below the explosive limit and to isolate the gas supply automatically in a fail-safe manner, without creating ignition sources or nuisance alarms.

B · CONTEXT

LPG leaks in kitchens cause fires and explosions every year, often when occupants are asleep, away or unaware of the smell. Low-cost metal-oxide sensors drift with age, humidity and temperature and can respond to cooking fumes, leading to false alarms or missed leaks. A more dependable system must combine sensing, self-diagnosis, safe actuation and clear alerts so the hazard is contained before a dangerous concentration builds up.

C · CORE FEATURES

- Continuously measure gas concentration, with warning and alarm levels defined as a percentage of the lower explosive limit (LEL).
- Sound a local alarm and send a remote alert when the alarm level is crossed.
- Close the gas supply automatically using a valve, and start ventilation using a spark-free method.
- Run self-diagnosis for sensor failure, warm-up, power loss and valve faults, with a fail-safe default.
- Require manual reset after a trip, and log events.

D · DESIGN REQUIREMENTS

- Users: homes, hostels, canteens and small restaurants.
- Safety first: fail-safe valve behaviour, no sparks or hot surfaces near the gas, conformity with applicable BIS/IEC safety guidance.
- Reliability: a justified balance between false alarms and missed detections, with a stated sensor life and calibration/replacement interval.
- Cost & resilience: affordable for households; continues to protect during a power cut.

E · TECHNICAL FOCUS

SENSING Metal-oxide, catalytic-bead or NDIR gas sensor with temperature and humidity compensation; discuss response time, cross-sensitivity and drift.

PROCESSING & CONTROL Microcontroller with warm-up handling, filtering, multi-level thresholds, drift compensation and watchdog; logic must default to the safe state on any fault.

ACTUATION Normally-closed solenoid or motorised valve, plus relay/driver design that avoids sparking near the gas line.

COMMUNICATION & POWER Local alarm plus optional Wi-Fi/GSM alert; mains supply with battery backup.

SIMULATION Mass-balance model of gas concentration in a kitchen with leak rate and ventilation, combined with a sensor-response model and the control logic; show detection and isolation time.

OPTIONAL Flame-failure sensing at the stove, CO sensing, regulator pressure-drop detection, voice alerts.

F · EXPECTED OUTCOME

- Block diagram and safety-oriented control flowchart.
- Simulation of at least three leak rates with concentration (% LEL) against time, alarm time and isolation time.
- Analysis of false-alarm and sensor-drift scenarios and how the design handles them.
- Fail-safe analysis showing the response to sensor, power and valve failure.
- Bill of materials, estimated cost and recommended maintenance schedule.

PS-04 DETECTION OF DRINKING WATER QUALITY CHANGES AT HOUSEHOLD SUPPLY POINTS

Recommended for Second & Third Year students

A · PROBLEM DEFINITION

Water reaching a household from the municipal network, a tanker or a borewell can change in quality because of pipe breaks, sediment in tanks, contamination or treatment failure. Residents have no quick way to notice such changes, and laboratory tests are slow and infrequent. The challenge is to detect deviations from normal water quality at the point of supply, using low-cost sensors, and to warn users before the water is consumed.

B · CONTEXT

Unsafe drinking water is a major cause of preventable illness, yet quality changes are often invisible until people fall sick. Periodic laboratory testing cannot catch short-lived events such as a turbidity spike after a pipe repair. A household-level device that tracks key indicators continuously and recognises unusual changes can give an early warning, prompt the use of alternate or treated water, and help communities report problems to the supplier.

C · CORE FEATURES

- Measure indicators such as pH, turbidity, total dissolved solids (TDS)/conductivity and temperature.
- Learn a baseline for the supply point and detect abnormal deviations or rapid changes.
- Classify status as normal, caution or unsafe using defined reference ranges.
- Warn residents locally and remotely, with an optional supply diversion or valve closure.
- Log quality trends over days and weeks.

D · DESIGN REQUIREMENTS

- Users: households, apartments, hostels and community water points.
- Cost & upkeep: affordable sensors with a stated calibration interval (pH electrodes drift), and low maintenance.
- Honest claims: the device is an indicator, not a laboratory-certified instrument; the documentation must say so.
- Reference limits: use accepted norms (e.g. IS 10500: pH 6.5–8.5, turbidity 1 NTU, TDS 500 mg/L); low false-alarm rate.

E · TECHNICAL FOCUS

SENSING pH, turbidity (optical), TDS/conductivity and temperature sensors; temperature compensation for pH and conductivity is expected.

PROCESSING & ANALYTICS Microcontroller with filtering, outlier rejection and a multi-parameter decision rule or composite water-quality index; justify sampling interval and thresholds.

COMMUNICATION Local indicator plus optional Wi-Fi, Bluetooth or GSM alerts.

POWER SUPPLY Mains or battery; consider flow-triggered sampling to save power and sensor wear.

SIMULATION Synthetic time-series of water parameters with injected events (turbidity spike, contamination, chlorine loss, sensor drift); show detection performance in Python, MATLAB or equivalent.

OPTIONAL Residual chlorine or ORP sensing, ML anomaly detection, tank-level monitoring, community-level dashboard.

F · EXPECTED OUTCOME

- System architecture and sensor-selection rationale.
- Simulation of normal operation and at least three contamination/change events with detection results.
- Classification logic with justified thresholds and example outputs.
- Discussion of calibration, drift and false-alarm management.
- Cost estimate and practical limitations compared with laboratory testing.

PS-05 AUTOMATIC BACKUP POWER MANAGEMENT FOR ESSENTIAL HOUSEHOLD APPLIANCES

Recommended for Third & Final Year students

A · PROBLEM DEFINITION

During power cuts, a home with a battery inverter, and possibly solar or a generator, must keep essential appliances (lights, fans, refrigerator, router, water pump) running for as long as possible. Fixed backup circuits either overload the battery or cut loads that matter most. The challenge is to design an automatic, multi-source backup manager that switches sources seamlessly and allocates limited stored energy intelligently among prioritised loads.

B · CONTEXT

Backup systems in homes are usually simple: the inverter feeds a fixed set of circuits until the battery is flat. Motor loads such as refrigerators and pumps draw several times their rated current at start-up, battery capacity falls with age, and the length of the outage is unknown. A system that estimates remaining runtime, sheds or defers low-priority loads, manages multiple sources and returns to the grid smoothly would extend critical-appliance uptime and protect the battery.

C · CORE FEATURES

- Detect grid failure, restoration and abnormal voltage/frequency, and transfer sources with minimal interruption.
- Arbitrate between grid, battery and, where available, solar or generator input.
- Estimate battery state of charge and remaining runtime at the present load.
- Prioritise loads and shed or defer non-essential ones automatically as the battery depletes.
- Handle motor start-up surges and protect against overload, short circuit and deep discharge.
- Show status, reserve time and event history to the user.

D · DESIGN REQUIREMENTS

- Users: households and small shops in areas with frequent outages.
- Value: longer uptime for refrigerator, communication and lighting loads than a fixed backup circuit.
- Battery life & cost: limit depth of discharge, charge current and temperature stress; affordable relative to the inverter.
- Safety & convenience: safe changeover (no grid back-feeding), simple priority configuration and clear indicators.

E · TECHNICAL FOCUS

SENSING Grid and inverter voltage/frequency, battery voltage, current and temperature, and branch-level load currents.

CONTROL Microcontroller or PLC-style logic. Define a priority scheme and a shedding rule; state changeover-time targets (e.g. below 20 ms for sensitive electronics, or a justified alternative).

ESTIMATION SoC estimation (coulomb counting, open-circuit voltage or a simple observer), plus an allowance for ageing and temperature.

POWER STAGE Relays, contactors, static switches or an inverter/charger modelled at system level; address interlocks that prevent two sources being connected together.

SIMULATION Time-domain simulation in Simulink, Python or equivalent covering outage events, motor start-up inrush, load shedding and battery depletion over several hours.

OPTIONAL Outage-duration prediction from history, solar and generator integration, appliance-level smart-socket control, tariff-aware recharge.

F · EXPECTED OUTCOME

- System architecture, source-transfer logic and load-priority scheme.
- Simulation of an outage showing changeover behaviour, load-shedding order and battery state against time.
- Quantified comparison with a fixed-circuit backup: critical-load uptime in hours for the same battery.
- Analysis of motor start-up inrush and how the design handles it.
- Cost estimate, failure-mode review and a note on safety compliance.